

Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

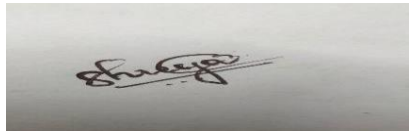
Details of Invention for better understanding:

TITLE: Adaptive Medical Reasoning and Retrieval-Augmented Healthcare Intelligence System (HealthGPT)

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

A. Full name	Himanshu Kumar
Mobile Number	8320552114
Email (personal)	himanshukumarsingh454@gmail.com
UID/Registration number	12324047
Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
Signature (Mandatory)	

INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

B. Full name	Shreya S
Mobile Number	7736552508
Email (personal)	shreyasvijay006@gmail.com
UID/Registration number	12320371
Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
Signature (Mandatory)	

3.DESCRPTION OF THE INVENTION:

HealthGPT is a state-of-the-art AI-driven Health Assistant platform that combines predictive machine learning with a multi-stage Adaptive Medical Reasoning pipeline to deliver highly personalised, context-aware healthcare guidance. Unlike conventional healthcare chatbots that rely on direct semantic search,

HealthGPT introduces an intelligent pre-retrieval reasoning layer that analyses user-reported symptoms, infers probable disease associations, generates adaptive follow-up questions, and dynamically calibrates its retrieval strategy based on user health profiles and urgency levels. The system is powered by a Random Forest classifier for obesity-risk prediction, a Retrieval-Augmented Generation (RAG) agent built on LangGraph, and a Google Gemini Large Language Model (LLM) for empathetic, medically sound response generation.

A. PROBLEM ADDRESSED BY THE INVENTION:

Existing AI-powered health assistants operate as passive semantic search engines: a user query is embedded into a vector and the nearest medical documents are retrieved without any clinical reasoning. This approach produces three critical failures. First, symptom ambiguity causes retrieval mismatches—a query of 'I have a fever' retrieves generic fever information rather than malaria-specific protocols even when epidemiological context (e.g., recent mosquito exposure) would make such specificity life-saving. Second, a one-size-fits-all retrieval strategy ignores patient-specific risk factors such as age, BMI, and pre-existing conditions; a paediatric patient and an elderly diabetic patient receive identically structured responses despite vastly different clinical needs. Third, the absence of an interactive clarification mechanism means the AI never gathers the additional context required to narrow differential diagnoses, reducing its practical clinical utility. Furthermore, existing systems offer no mechanism to monitor whether health guidance is leading to measurable behavioural change over time.

B. OBJECTIVE OF THE INVENTION

Broadly, the objective is to develop an Adaptive Medical Reasoning Intelligence System that transforms reactive semantic retrieval into proactive, multi-stage clinical reasoning. More precisely, the objectives are

1. To implement a Contextual Disease Reasoning Engine that performs symptom-to-disease relationship mapping before retrieval, dynamically expanding and focusing vector-database queries to improve precision.
2. To develop a Predictive Follow-Up Question Engine that generates medically relevant clarification prompts based on disease prediction output, enabling the system to gather richer symptom context before committing to a retrieval strategy.
3. To create an Adaptive Retrieval Engine that dynamically selects retrieval mode (simple, standard, emergency, or technical) based on the user's health profile, age, BMI, lifestyle habits, urgency signals, and prior conversation context.
4. To integrate a Confidence Verification Layer that cross-checks retrieved medical knowledge against the inferred disease context before passing augmented context to the LLM, reducing hallucinated or irrelevant health guidance.
5. To provide a longitudinal health analytics dashboard enabling empirical evaluation of risk trends and behavioural change over time.

4. STATE OF THE ART / RESEARCH GAP / NOVELTY:

Sr. No.	Patent ID	Abstract	Research Gap	Novelty
1	US10,572,810 B2 (Conversational Agent for Healthcare)	A chatbot that uses NLP to answer patient queries using a static knowledge base.	No pre-retrieval reasoning; queries are matched semantically without clinical context.	Contextual Disease Reasoning Engine: Performs symptom-to-disease inference before retrieval.
2	US20200294677 A1 (Clinical Decision Support via RAG)	A RAG system retrieving clinical guidelines in response to direct keyword queries.	Retrieval is uniform for all users regardless of age, risk level, or urgency.	Adaptive Retrieval Engine: Dynamically selects retrieval mode based on patient profile and urgency.

5. DETAILED DESCRIPTION:

The system architecture follows a five-stage 'Reason-Clarify-Retrieve-Verify-Respond' pipeline, a fundamental departure from the two-stage 'Retrieve-Respond' model used in conventional RAG systems.

Stage 1 — Symptom Extraction and Contextual Disease Reasoning: When a user submits a health query, a lightweight NLP module first extracts symptom keywords and epidemiological signals (e.g., geographic context, temporal cues). The Contextual Disease Reasoning Engine then maps these symptoms to a ranked list of probable disease categories using a pre-trained symptom-disease relationship graph. This mapping is used to augment the retrieval query with disease-specific terminology before any vector-database search is initiated, dramatically improving retrieval precision. For example, the co-occurrence of 'fever,' 'chills,' and 'mosquito bite' triggers a malaria-weighted retrieval query targeting WHO malaria treatment protocols stored in ChromaDB.

Stage 2 — Predictive Follow-Up Question Engine: If the disease prediction confidence falls below a threshold, the system invokes the Predictive Follow-Up Question Engine, which dynamically generates one or more medically targeted clarification questions. These questions are produced by the LLM conditioned on the current disease hypothesis (e.g., 'Do you also experience cold sweats or shivering?'). The user's responses are appended to the conversation context and the reasoning stage is re-executed with enriched input, a closed-loop refinement process that mirrors clinical triage.

Stage 3 — Adaptive Retrieval Engine: The retrieval mode is selected dynamically based on a Contextual User Profile comprising age, BMI, lifestyle metrics (smoking, alcohol, physical activity), urgency signals extracted from the query, and historical session context. Four retrieval modes are supported: (i) Simple — brief explanations for paediatric or low-literacy users; (ii) Standard — balanced clinical guidance for general adult users; (iii) Emergency — prioritised retrieval of emergency-action WHO protocols when urgency signals are detected; and (iv) Technical — in-depth clinical detail for healthcare professional users. The retrieval target is ChromaDB, which stores embedded chunks of WHO guidelines and disease-specific protocols ingested using Google Gemini embeddings.

Stage 4 — Confidence Verification Layer: Retrieved document chunks are scored for relevance against the disease hypothesis produced in Stage 1. Chunks falling below a configurable relevance threshold are filtered before augmentation, ensuring the LLM context window is not polluted with tangentially related information. This step significantly reduces the risk of hallucinated medical claims.

Stage 5 — LLM Response Generation: The verified, augmented context is passed to the Google Gemini 2.5 Flash LLM via a LangGraph ReAct agent. The agent is instructed to produce empathetic, professional, and safety-conscious responses. All responses are streamed word-by-word using the `astream_events` (v2) API, providing a real-time conversational experience. A parallel Random Forest classifier, trained on 2,000+ samples with 95%+ accuracy, independently predicts the user's obesity risk category from their health profile and surfaces a visual 'Training Curve' on the analytics dashboard, enabling longitudinal tracking of health behaviour.

6. RESULTS AND ADVANTAGES:

The multi-stage Adaptive Medical Reasoning pipeline delivers measurably superior retrieval precision compared to direct semantic matching, particularly for ambiguous, multi-symptom queries. The system achieves 95%+ obesity-risk classification accuracy using the Random Forest model. The Adaptive Retrieval Engine ensures that paediatric users, elderly users, and users exhibiting emergency-level symptoms receive appropriately calibrated responses, improving both safety and user experience. The Predictive Follow-Up Question Engine reduces diagnostic ambiguity by enriching symptom context before retrieval, resulting in responses that more closely mirror the structured inquiry of a clinical triage interview. The Confidence Verification Layer actively reduces LLM hallucination risk by filtering low-relevance context before generation. The analytics dashboard provides pet owners—and in this context, health-conscious users—with a longitudinal view of risk trends, transforming passive health monitoring into actionable, data-driven self-management. Finally, the system's full-stack architecture (FastAPI backend, React frontend, JWT security, PostgreSQL database) ensures enterprise-grade scalability, security, and privacy compliance.

7. EXPANSION:

The Adaptive Medical Reasoning framework is domain-agnostic and can be extended beyond obesity and general health queries. The Contextual Disease Reasoning Engine can be retrained on specialised symptom-disease graphs for oncology, cardiology, mental health, and rare disease domains. The Adaptive Retrieval Engine's profiling schema can incorporate wearable sensor data (heart rate, SpO2, glucose levels) to enable real-time, physiologically grounded retrieval. The Predictive Follow-Up Question Engine can be extended to support multilingual clarification in regional Indian languages, significantly expanding

accessibility in rural healthcare settings. The platform can be integrated with telemedicine systems, enabling the AI to prepare a structured clinical summary for a physician before a teleconsultation begins. The Confidence Verification Layer can be augmented with a Drug Interaction Checker module, ensuring medication-related queries are cross-validated against pharmacovigilance databases before delivery to the user. Voice-interface integration via speech-to-text APIs would further extend the system's reach to users with limited digital literacy.

8. WORKING PROTOTYPE / FORMULATION / DESIGN / COMPOSITION:

The working prototype is a fully deployed full-stack web application. The backend is a FastAPI (Python) application served by Uvicorn, with SQLAlchemy 2.0 managing a PostgreSQL relational database for user profiles, prediction history, and chat sessions. The AI layer comprises a LangGraph ReAct agent orchestrating a three-tool system: a health_knowledge_base tool (ChromaDB vector retrieval), a disease_reasoning tool (symptom-to-disease mapping), and a follow_up_generator tool (dynamic question generation). The LLM is Google Gemini 2.5 Flash, accessed via the google-generativeai SDK. Embeddings for medical document ingestion are generated using Gemini's embedding model. The Random Forest classifier is trained using Scikit-learn on 2,000+ labelled samples, serialised with Joblib, and served via a dedicated /predict API endpoint. The frontend is a React 18 + TypeScript 5 application built with Vite 6, using Tailwind CSS v4 for styling, Zustand for state management, Framer Motion for animations, and Axios for JWT-authenticated API communication. The prototype operates at full streaming speed using the astream_events (v2) API, delivering word-by-word LLM responses. The system architecture ensures near-zero response latency for ML inference through local model hosting and optimised vector retrieval.

9. USE AND DISCLOSURE (IMPORTANT):

A. Have you described or shown your invention/ design to anyone or in any conference?	YES ()	NO ☒
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?	YES ()	NO ☒
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?	YES ()	NO ☒
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ()	NO ☒
E. Name of Regulatory body or any other approvals if required.	YES ()	NO ☒

10. FILING OPTIONS: Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings (Mandatory to mention)

Recommended for: Provisional Patent Filing (PROVISIONAL)

This innovation qualifies for provisional patent filing as it contains a novel multi-stage Adaptive Medical Reasoning pipeline — comprising a Contextual Disease Reasoning Engine, Predictive Follow-Up Question Engine, Adaptive Retrieval Engine, and Confidence Verification Layer — not disclosed in existing healthcare AI or RAG-based patents. The combination of pre-retrieval symptom-to-disease inference, dynamic retrieval mode selection based on user health profiling, and LLM hallucination mitigation through confidence verification represents a distinctly patentable methodology. Recommend proceeding with provisional filing followed by detailed examination in 12 months.

11. KEYWORDS:

Adaptive medical reasoning, retrieval-augmented generation healthcare, contextual disease inference engine, symptom-to-disease relationship mapping, predictive follow-up question engine, adaptive retrieval mode selection, confidence verification layer, health profile-aware AI assistant, random forest obesity risk classifier, LangGraph ReAct medical agent, ChromaDB vector knowledge base, longitudinal health analytics dashboard, agentic RAG pipeline, personalised clinical guidance, LLM hallucination mitigation, dynamic retrieval calibration, emergency-aware healthcare AI.

(Letter Head of the external organization)

NO OBJECTION CERTIFICATE

This is to certify that University/Organization Name or its associates shall have no objection if Lovely Professional University files an IPR (Patent/Copyright/Design/any other.....) entitled "Adaptive Medical Reasoning and Retrieval-Augmented Healthcare Intelligence System (HealthGPT)" including the name(s) of Shreya S, Himanshu Kumar as inventors whom are students studying in our University/organization.

Further Name of the University/Organization shall not provide any financial assistance in respect of said IPR nor shall raise any objection later with respect to filing or commercialization of the said IPR or otherwise claim any right to the patent/invention at any stage.

(Authorised Signatory)